

Lab # 05

Retrieval of Data from Relational Database Systems using Query Language

Objective:

Student will learn about to do the following:

- Select specific data through WHERE clause
- Operator allowed in WHERE clause

Software Tools:

- MySQL Workbench

Theory:

The WHERE clause is used to filter records. The WHERE clause is used to extract only those records that fulfill a specified condition.

WHERE Syntax

```
SELECT column1, column2,...
FROM table_name
WHERE condition;
```

Example

```
SELECT * FROM Customers WHERE
Country='Mexico';
```

Quotes Around Text Fields

SQL uses single quotes around text values (most database systems will also accept double quotes).

Although, numeric values should not be enclosed in quotes.

For text values:

This is correct:

```
SELECT * FROM Persons WHERE FirstName='Tove';
```

This is wrong:

```
SELECT * FROM Persons WHERE FirstName=Tove;
```

For numeric values:

This is correct:

SELECT * FROM Persons WHERE Year=1965;

This is wrong:

SELECT * FROM Persons WHERE Year='1965';

Operators Allowed in the WHERE Clause

With the WHERE clause, the following operators can be used:

Operator	Description	Example
=	Equal	WHERE Price = 18;
<>	Not equal	WHERE Price <> 18;
>	Greater than	WHERE Price > 18;
<	Less than	WHERE Price < 18;
>=	Greater than or equal	WHERE Price >= 18;
<=	Less than or equal	WHERE Price <= 18;
BETWEEN	Between an inclusive range	WHERE Price BETWEEN 18 AND 50;
LIKE	Search for a pattern	WHERE City LIKE 's%';
IN	If you know the exact value you want to return for at least one of the columns	WHERE City IN ('Paris', 'London');

Note: In some versions of SQL, the <> operator may be written as !=

The AND & OR Operators

The AND & OR operators are used to filter records based on more than one condition. The AND operator displays a record if both the first condition and the second condition are true. The OR operator displays a record if either the first condition or the second condition is true.

AND Operator Example

The "Persons" table:

P_Id	LastName	FirstName	Address	City
1	Hansen	Christ	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes
3	Pettersen	Michael	Storgt 20	Stavanger

Now we want to select only the persons with the first name equal to "Tove" AND the last name equal to "Svendson":

We use the following SELECT statement:

```
SELECT * FROM Persons
WHERE FirstName='Tove'
AND LastName='Svendson' ;
```

The result-set will look like this:

P_Id	LastName	FirstName	Address	City
2	Svendson	Tove	Borgvn 23	Sandnes

OR Operator Example

Now we want to select only the persons with the first name equal to "Tove" OR the first name equal to "Christ":

We use the following SELECT statement:

```
SELECT * FROM Persons
WHERE FirstName='Tove'
OR FirstName='Christ'
```

The result-set will look like this:

P_Id	LastName	FirstName	Address	City
1	Hansen	Christ	Timoteivn 10	Sandnes
2	Svendson	Tove	Borgvn 23	Sandnes

Combining AND & OR

You can also combine AND and OR (use parenthesis to form complex expressions).

Now we want to select only the persons with the last name equal to "Svendson" AND the first name equal to "Tove" OR to "Christ":

We use the following SELECT statement:

```
SELECT * FROM Persons
WHERE LastName='Svendson'
AND (FirstName='Tove' OR
     FirstName='Christ')
```

The result-set will look like this:

P_Id	LastName	FirstName	Address	City
2	Svendson	Tove	Borgvn 23	Sandnes

IN Operator

It allows you to specify multiple values in a WHERE clause.

```
SELECT column_name(s)
FROM table_name
WHERE column_name IN (value1,value2,...);
```

P_Id	LastName	FirstName	Address	City
1	Hansen	Christ	Timofeev 10	Atlanta
2	Svendson	Tove	Burgan 23	Sandnes
3	Pettersen	Michael	Sturt 20	Stavanger
4	Pettersen	Christ	Borgne 24	Atlanta
5	Calvin	Harry	Borgne 27	Miami

```
SELECT * FROM Persons
WHERE City IN ('Miami', 'Atlanta');
```

The result-set will look like this:

P_Id	LastName	FirstName	Address	City
1	Hansen	Christ	Timofeev 10	Atlanta
4	Pettersen	Christ	Borgne 24	Atlanta
5	Calvin	Harry	Borgne 27	Miami

BETWEEN Operator

The BETWEEN operator selects values within a range. The values can be numbers, text, or dates.

```
SELECT column_name(s)
FROM table_name
WHERE column_name BETWEEN value1 AND value2;
```

Example:

Order_Id	LastName	FirstName	Address	City	OrderPrice
1	Hansen	Christ	Timofeev 10	Atlanta	1000
2	Svendson	Tove	Burgan 23	Sandnes	550
3	Pettersen	Michael	Sturt 20	Stavanger	4677
4	Pettersen	Christ	Borgne 24	Atlanta	234
5	Calvin	Harry	Borgne 27	Miami	8000

```
SELECT * FROM Customers
WHERE OrderPrice BETWEEN 1000 AND 5000;
```

The result-set will look like this:

Order_Id	LastName	FirstName	Address	City	OrderPrice
1	Hansen	Christ	Timofeev 10	Atlanta	1100
3	Pettersen	Michael	Sturt 20	Stavanger	4677

NOT Operator

The WHERE clause can be combined with NOT operators. The NOT operator displays a record if the condition(s) is NOT TRUE.

SELECT *column1, column2, ...* **FROM** *table_name*
WHERE NOT *condition*;

Example:

Cust_Id	LastName	FirstName	Address	Country
1	Hansen	Christ	Timofeev 10	Australia
2	Svendson	Tove	Burgan 23	Sydney
3	Pettersen	Michael	Sturt 20	England
4	Pettersen	Christ	Borgne 24	Sweden
5	Calvin	Harry	Borgne 27	Germany

```
SELECT Country FROM Customers  
WHERE NOT Country='Germany';
```

The result-set will look like this:

Country
Australia
Sydney
England
Sweden

Combining AND, OR and NOT

You can also combine the AND, OR and NOT operators.

```
SELECT * FROM Customers  
WHERE Country='Germany' AND (City='Berlin' OR City='München');
```

```
SELECT * FROM Customers  
WHERE NOT Country='Germany' AND NOT Country='USA';
```

LAB TASK

Create a table Employee having primary key with the following schema:

(Emp_no, E_name, E_city, Dept_ID, Dept_name, Dept_HOD, Salary, E_joiningdate, E_serviceyear). Insert at least 5 rows in the table:

Write SQL query for the following questions

1. Select all department name and respective HOD name from Employee table.
2. Display all information of the Employee table.
3. Display all employee whose joining date is from 1st January to 30th September.
4. Select employee with emp_id greater than 10 or dept_ID equal to 1.
5. Select salary where it can be 10,000, 25,000 and 50,000.
6. Display the complete record of employees with salary less than 25,000 and city Taxila.

Conclusion:
